

Hint for Q2

Friday, 11 September 2026

00:35

To deal with applying recursive definition to $\mathbb{N} \times \mathbb{N}$, consider following.

Recall recursive definition:

Given $c \in \mathbb{N}$, $f_n: \mathbb{N} \rightarrow \mathbb{N}$

$$a_0 = c, \quad a_{n++} = f_n(a_n).$$

Construction 1: Let $f_n = \phi$, where

$\phi: \mathbb{N} \rightarrow \mathbb{N}$ is defined as

$$\phi(y) = 1 \quad \text{if } y = 0$$

$$\phi(y) = 0 \quad \text{if } y \neq 0.$$

Then define

$$p_0 = 0, \quad p_{n++} = \phi(p_n).$$

(This is valid
recursion)

Construction 2: Define

$f_n: \mathbb{N} \rightarrow \mathbb{N}$ as

$$f_n(y) = y \quad \text{if } p_n = 0$$

$$f_n(y) = y_{++} \quad \text{if } p_n \neq 0$$

Define $a_0 = 0$, $a_{n++} = f_n(a_n)$

Then set $h(n) = a_n$.

Verify that $h(0) = 0$

$$h(0_{++}) = 0$$

$$h(n_{++}) = h(n)_{++}.$$